

[DG] General AI Fluency — Impact Project Capstone

Intern: Divine Gabriel | Track: Backend AI Engineering (Cybersecurity & Backend)

1. Portfolio Architecture & Case Study Destination

Future case studies, security reports, and backend builds will be published directly to the active GitHub Pages portfolio repository (flyrank-portfolio) deployed at <https://gabydidi.github.io/flyrank-portfolio/>. All new additions will strictly adhere to the established brand identity kit: Deep Slate background (#0F172A), Soft Off-White text (#F1F5F9), and Emerald Green accents (#10B981).

2. The Standardized 3-Beat Content Shape

To ensure consistency across future portfolio expansions, every technical case study will follow a rigorous 3-beat structure:

- **The Problem:** A clear technical challenge, infrastructure vulnerability, or backend routing objective faced during development or penetration testing labs (drawing from practical experience at Ciphernet Technologies).
- **What You Did:** The concrete implementation steps taken, including Python scripting, network configuration, API endpoint design, or vulnerability exploitation methodology using industry tools (Nmap, Metasploit, Burp Suite, Flask).
- **What Came of It:** The verified outcome, successful deployment proof, captured telemetry, or remediated security posture.

3. Named Next Piece of Work

The immediate next case study queued for publication is a Python-Based Automated TCP Port Scanner with Vulnerability Fingerprinting and JSON Reporting, integrating socket programming and multi-threaded scanning techniques.

4. Maintenance & Reminder Evidence

To prevent portfolio stagnation and maintain professional momentum:

- **Reminder Schedule:** A recurring calendar nudge set for every Monday at 9:00 AM labeled "Portfolio Case Study & SIWES Report Review".
- **Context Preservation:** The entire build architecture, style guidelines, and technical scope are

permanently indexed within this persistent Claude Project environment, ensuring future content updates remain streamlined and low-friction.

Submitted in partial fulfillment of the FlyRank AI Internship – General AI Fluency Track.